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First-principles calculations reveal controlling principles for carrier mobilities in semiconductors

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Abstract

Carrier mobilities remain a key qualifying factor for materials competing for next-generation electronics. It has long been believed that carrier mobilities can be calculated using the Born approximation. Here, we introduce a parameter-free, first-principles approach based on complex-wavevector energy bands which does not invoke the Born expansion. We demonstrate that phonon-limited mobility is controlled by low-resistivity percolation paths, which arise from fluctuations that are beyond the Born approximation. We further demonstrate that, in ionized-impurity scattering, one must account for the effect of the screening charge, which cancels most of the Coulomb tail. Calculated electron mobilities in silicon are in agreement with experimental data. The method is easy to use and can provide guidance in the search for high-mobility device designs.

Keywords: semiconductors, electron scattering, carrier mobility

(Some figures may appear in colour only in the online journal)

Carrier mobility is the most important quantity that controls the performance of electronic devices. It is an old and respected area of research, but it has acquired renewed significance in recent years for several reasons. An intense search is under way to obtain enhanced mobilities in Si-based electronics in order to extend Moore's law. In addition, there is extensive research to design devices using two-dimensional materials, ultimately aiming for ultrathin flexible electronics. More specifically, a promising way to achieve higher mobility in Si-based MOSFETs (metal–oxide–semiconductor field-effect transistors) is to line the channel with high-mobility III-V semiconductors [1], or with other high-mobility materials such as carbon nanotubes or graphene [2, 3]. In such configurations, however, mobilities are affected by the interfaces on both sides of the new layer. Free-standing monolayer graphene is renowned for its ultrahigh carrier mobilities at low temperatures [4], but phonons cause significant mobility degradation at room temperature [5]. Similarly, carrier mobility degrades significantly when graphene is placed on a substrate. Mobility issues also arise in topological insulators. For example, it has been found that, in topological insulators with ultrahigh bulk mobilities, the surface mobility is reduced significantly despite its topological protection [6].

Device and circuit engineers rely on mobility models, which ultimately must be based on fundamental physics. Rigorous quantum mechanical frameworks exist [7–9] for calculating effects of impurity and defect scattering, but practical implementations of these schemes are hindered by prohibitive computation resource requirements [10]. A few calculations based on first-principles density-functional theory (DFT) calculations have relied on scattering rates calculated using a formula in a form derived from the Born approximation [11–14]. The accuracy of this approximation is not known. In Si electronics, engineering-level device modeling has traditionally relied on the same approximation, using phenomenological descriptions of independent scattering mechanisms

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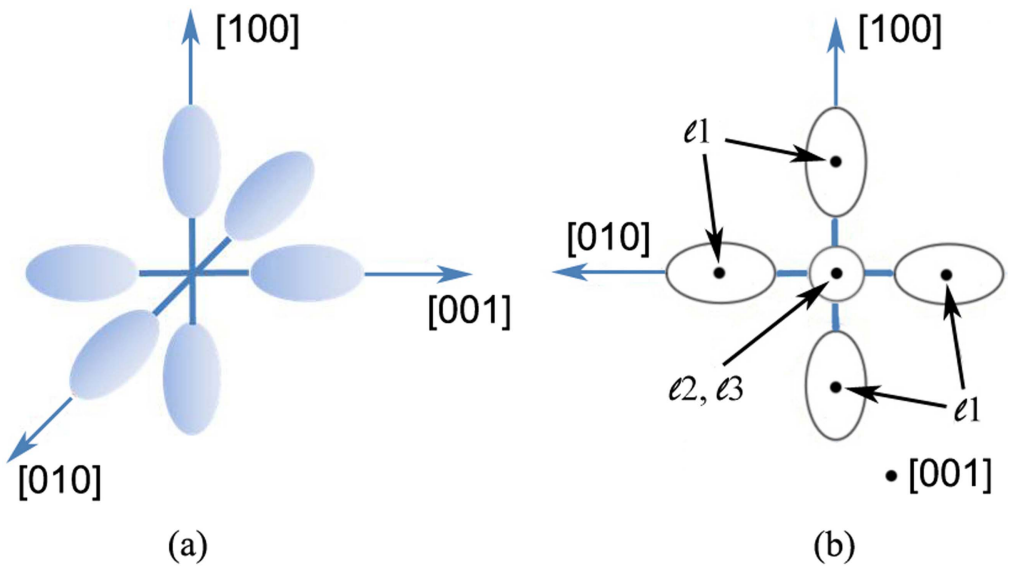

Figure 1. (a) The energy isosurface (ellipsoids) of Si above conduction-band minimum. The current flows along the (001) direction. (b) The isosurface viewed from the (001) direction.

(phonon scattering, Coulomb scattering by ionized impurities, scattering by interface roughness, etc) [15]. Even though the approach entails free parameters that are fitted to data, it is not always possible to do so with physically meaningful parameters, as for example in ultrathin silicon-on-insulator structures and in strained Si channels [16, 17]. Overall, the success of the Born approximations in engineering-level models that entail fitting parameters does not validate the approximation. As big-data simulations of materials by design is picking up momentum, a reliable, parameter-free method for calculating mobilities is needed.

In this paper, we demonstrate that the Born approximation, which is at the core of all engineering-level mobility modeling and DFT-based calculations, provides an incorrect description of the effect of phonons on carrier mobility for physically transparent reasons. The Born approximation views the electron as traveling in a solid in which the number of phonons in each mode is equal to the mean, typically non-integer value dictated by Bose–Einstein statistics as a function of temperature. A carrier scatters independently from each of these phonons to linear order and the scattering rates are summed up. In contrast, we demonstrate that fluctuations that correspond to higher-order terms in the Born expansion lead to the formation of percolation 'paths of least resistance', determining the value of mobility. We arrive at this result by introducing an altogether different approach for parameter-free, DFT calculations of carrier mobilities. The method does not rely on the Born expansion. Instead, we recognize that the mean lifetime of a carrier (the inverse of the mean scattering rate) in the presence of scatterers can be represented by an imaginary component in the carrier's energy. We then show that one can equivalently describe the carrier as having a real energy, but a complex wavevector. For a single scattering center or a particular configuration of scatterers within a supercell, this equivalence is achieved by constructing an equivalent supercell containing a homogeneous effective medium with a complex potential to represent the effect of the scatterers. We invoke the well-established theory of complex-wavevector energy bands (complex bands) and produce an algorithm that yields mean lifetimes that are then inserted in the linearized Boltzmann equation to produce mobilities. Multiple configurations of scatterers can be introduced to capture the effect of spatial inhomogeneities introduced by fluctuations. We implement the method for phonon scattering in Si and indeed find that the mobility is controlled by percolation paths of least resistance and obtain excellent agreement with experimental data. We also apply the same method to ionized-impurity scattering and demonstrate that the usual practice of determining the scattering rate for a single impurity with a Coulombic potential (infinite $1/r$ tail) and multiplying by the density of impurities is not at all appropriate. In contrast, we show that the Coulomb tail vanishes at half the inter-impurity separation, and this feature is an essential ingredient for the calculation of the resulting mobility. Once more, agreement with the data for impurities in Si validates our conclusion. The method is easy to implement and can be used for reliable calculations of mobilities in any system.

The focus of the present theory is the elastic scattering time τ of carriers, which is equivalent to the momentum relaxation time for isotropic scatterers. This approximation [18] has been used in all prior theories for mobility and the present theory does not go beyond it. In the present work, we further show that ionized impurities do not have long-range Coulombic tails, which would induce small-angle scattering. The relaxation time is a function of the electron wavevector $\mathbf{k}$. At relatively low temperatures, $\mathbf{k}$ is confined in the vicinity of the band minimum or minima. If z is the direction of the current, it is convenient to define $\tau_\ell(k_x, k_y, \epsilon)$, where ϵ is the carrier's energy and ℓ is the valley index. Here, we replace k_z with ϵ as the independent variable, but the two $\pm k_z$ values correspond to the same ϵ. Thus, we extend the valley index ℓ to go over the two k_z branches. More specifically, in Si, which has six equivalent conduction-band minima, there are twelve

ℓ values (as shown in figure 1.) We show in appendix A that, in the linearized Boltzmann theory, after integration over the carrier energy ϵ and replacing the integration over (k_x, k_y) by a sum over the points $(k_{x\ell}, k_{y\ell})$, each of which corresponds to the location of the band bottom of a valley, the mobility can be expressed in the simple form,

$$\mu = \frac{e}{12}\sum_{\ell}\frac{\tau_\ell\left(k_{x\ell}, k_{y\ell}, \frac{1}{2}k_B T\right)}{m_{z,\ell}}. \tag{1}$$

Here, $m_{z,\ell}$ is the effective mass in the z direction in the ℓth valley. Although there are a total of 12 k points in this sum, there are only three inequivalent ones due to symmetry. As shown in figure 1, assuming the current flows along the [001] direction, the electron can either be on the transverse ellipsoids (denoted as $\ell 1$), or on the longitudinal ellipsoids (denoted as $\ell 2$ and $\ell 3$). The remaining task is to calculate the scattering lifetime τ_ℓ.

In the conventional approach to mobility calculations, the lifetimes τ_ℓ in equation (1) are calculated by perturbation theory, which requires the construction of scattering potentials, proper inclusion of electronic screening, and calculation of the pertinent transition matrix elements. Such calculations require accurate treatment of electronic screening, which in some cases requires the development of elaborate schemes [19]. The method we introduce in this paper avoids all difficulties associated with constructing scattering potentials, electronic screening, and matrix elements because it is not based on perturbation theory. Instead, it constructs an effective medium by requiring that it has the same electron transport properties (i.e. mobility) as the real medium. According to the Landauer formula [20, 21], the total current is determined by the transmission probability of the electron wave function through the scattering region. Thus, the effective medium should be a homogeneous medium that yields the same transmission probability as the real, inhomogeneous medium. A previous work [22] connected the transmission probability of a junction with the complex-wavevector bands of a supercell with absorbing boundary conditions. This work relies on a similar connection, i.e. in a homogeneous effective medium that represents ensemble-averaged defect-scattering effects, all Bloch waves become decaying waves, and their decay rates determine the transmission probability and transport properties. One way to utilize this connection is to introduce a complex constant potential in the form of an imaginary self-energy, as we will show below. This construction allows the inclusion of all electronic screening effects through DFT self-consistent calculations of both the real material and the effective medium.

It has been shown by Altshuler [23] that, for scattering due to uncorrelated deviations from crystallinity (defects, impurities, atomic displacements), an effective medium obtained from an appropriate configurational average acquires an imaginary 'self-energy' to that related to the mean scattering lifetime τ. In our case this relationship is

$$\mathrm{Im}\,\Sigma_\ell(\mathbf{k}, \epsilon) = \frac{\hbar}{2\tau_\ell(\mathbf{k}, \epsilon)}, \tag{2}$$

where $\mathbf{k} = (k_x, k_y)$, and the factor 2 accounts for the difference between wave function and density lifetimes. The complex self-energy $\Sigma_\ell(\mathbf{k}, \epsilon)$ for the effective medium is normally calculated via a configurational average over the random deviations from crystallinity, usually a daunting task, or through a mean-field theory such as the coherent potential approximation (CPA) [24, 25]. Such an approach, however, is not available to plane-wave-based first-principles methods. We will pursue an alternative approach. As discussed above, the complex self-energy for each Bloch state must be such that a supercell of the effective medium yields the same transmission probability for the Bloch electrons in the direction of the current as that from the scattering in a corresponding supercell containing scatterers in the real crystal. This condition turns the problem of finding the self-energy of the effective medium into that of a complex-wavevector band structure calculation.

A final step in our method is to remove the interference between the scattered waves from neighboring supercells. Luttinger and Kohn [7] showed that an ensemble average of random impurities removes the influence of impurity positions. In the present approach, an imaginary potential produces a mean-field effective medium that is equivalent to an ensemble average. We accomplish the task by adding a constant imaginary part ($i\eta$) to the self-energies of both the real configuration and the effective medium. Adding $i\eta$ in the real configuration can accomplish the desired goal, but it may also introduce changes to the band dispersion. By also adding $i\eta$ to the effective medium, we ensure that both systems are changed in approximately the same way. We also keep η as small as possible to minimize possible errors (see below). With this setup, we require that the effective medium yields the same transmission probability along the current direction as the real configuration. Because $\mathrm{Im}\,k_z$ uniquely determines the transmission probability [22], the requirement that the real configuration and the effective medium yield the same transmission is equivalent to equating the imaginary part of k_z in the two systems,

$$\mathrm{Im}\,k_z(i\eta) = \mathrm{Im}\,k_z^{em}(i\,\mathrm{Im}\,\Sigma + i\eta), \tag{3}$$

where the parentheses denote functional dependence, k_z is the complex wavevector for the real configuration, and k_z^{em} is that for the effective medium. Equation (3) is the central result of this paper. It defines the effective medium in terms of its complex-wavevector band structure, which yields the same transmission probability as that of the real medium.

To solve for $\mathrm{Im}\,\Sigma(\mathbf{k}, \epsilon)$ from equation (3), we iterate the following Newton–Raphson equation,

$$i\,\mathrm{Im}\,\Sigma^{(n+1)} = i\,\mathrm{Im}\,\Sigma^{(n)} + \frac{\mathrm{Im}\,k_z(i\eta) - \mathrm{Im}\,k_z^{em}(i\,\mathrm{Im}\,\Sigma^{(n)} + i\eta)}{d\,\mathrm{Im}\,k_z^{em}/d\Sigma} \tag{4}$$

until $\mathrm{Im}\,\Sigma^{(n)}$ converges, where $d\,\mathrm{Im}\,k_z^{em}/d\Sigma$ is the complex slope of $\mathrm{Im}\,k_z^{em}$ as a function of $\Sigma(\mathbf{k}, \epsilon)$. Note that the focus of this procedure is the lifetime of the electron caused by scattering out of its Bloch state, without tracking where it goes. The final $\mathbf{k}'$ value does not explicitly enter the

equations, whereby both intra-valley and inter-valley scattering is automatically included.

The minimum value for η that can remove interference between neighboring supercells is estimated from the relationship between the effective lifetime due to η, $\tau = \hbar/\eta$ and the corresponding mean-free-path, $\lambda = |\mathbf{v}_g|\tau$, where $\mathbf{v}_g = (1/\hbar)\partial\epsilon/\partial\mathbf{k}$ is the group velocity. Equating λ with half of the supercell edge, $L/2$, we find, $\eta = 2\hbar|\mathbf{v}_g|/L$. η can be viewed as a convergence parameter. As the size of the supercell is increased, η is reduced accordingly. It decreases to zero as the size of the supercell goes to infinity, thus achieving convergence. For a fixed impurity concentration, large supercells would contain several impurities which would necessitate a configurational average. For phonon scattering, the distance between atoms in neighboring supercells with the same displacement is also L so the same scaling of η is appropriate.

We have applied the above methodology to mobilities limited by phonon scattering and by ionized impurities. The calculations are carried out using DFT as implemented in the Quantum Espresso package [26] for the electronic structure part, and the PWCOND [27, 28] code to calculate the complex band structure. The latter is modified to allow an additional constant imaginary potential, which we set to be the negative of equation (4). For ionized-impurity scattering, 216-atom and 512-atom cubic supercells are used, each containing a single charged impurity atom, i.e. corresponding to impurity concentrations equal to $2.26 \times 10^{20}\,\mathrm{cm}^{-3}$ and $9.53 \times 10^{19}\,\mathrm{cm}^{-3}$, respectively. The phonon modes are calculated using density-functional perturbation theory (DFPT) [29, 30].

To compute the phonon-limited mobility, a set of frozen phonon configurations are generated from calculated phonon modes with the displacement for each atom determined by the requirement that the phonon modes obey the Bose–Einstein distribution,

$$\mathbf{u}_n = \sum_{\mathbf{q},i} \mathbf{u}_{n,\mathbf{q}i} = \sqrt{\frac{\hbar}{m_n}} \sum_{\mathbf{q},i} \frac{\hat{n}_{\mathbf{q}i} e^{i(\mathbf{q}\cdot\mathbf{R}_n + \phi_{\mathbf{q}i})}}{\sqrt{(e^{\hbar\omega_{\mathbf{q}i}/k_B T} - 1)\omega_{\mathbf{q}i}}}, \qquad (5)$$

where m_n is the mass of atom n, $\mathbf{q}$, i are mode indices, and $\hat{n}_{\mathbf{q}i}$ is the eigenvector for each mode normalized over the supercell, $\sum_n |\hat{n}_{\mathbf{q}i}|^2 = 1$. Each mode is added with a randomized phase $\phi_{\mathbf{q}i}$ to remove correlation between different modes. We first use a 2000 atom supercell to generate the frozen phonon configurations and then take a 512-atom section to form the supercell for the mobility calculation. An important aspect of phonon scattering is that the incident electron should have a real wavevector k_z that contains the average effect of all phonon configurations (there is no 'pristine' phonon-free medium). To reflect this feature in an approximate manner, for each configuration, the electrostatic potential of the effective medium is shifted so that it yields the same real part of complex k_z as the frozen phonon configuration. The final step is to determine the imaginary self-energy from the complex bands using equation (4). Note that, here, the ensemble average over scattered electronic states, to remove the phase coherence and yield the effective medium, is performed for each phonon configuration. The sum over the phonon configurations is performed once the imaginary self-energy is found for each configuration to account for non-linear phonon effects that are not included in a conventional single phonon–electron scattering theory [31].

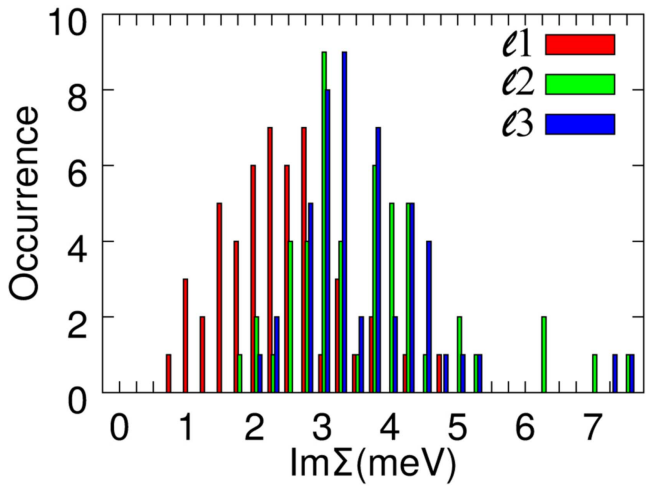

Figure 2. Distribution of Im Σ for 50 phonon configurations. Red, green and blue bars represent contributions from $\ell 1$, $\ell 2$ and $\ell 3$ valleys in the Si conduction band, respectively.

Figure 2 shows the statistics of the calculated for configurations at 300 K. Compared to electrons at $\ell 2$ and $\ell 3$, electrons at $\ell 1$ (transverse ellipsoids) have relatively smaller imaginary energies, or longer lifetimes. Electrons on the transverse ellipsoids also have a much smaller effective mass than electrons on the longitudinal ellipsoids ($\ell 2$ and $\ell 3$), and thus larger velocities. Moreover, eight out of twelve terms in the sum of equation (A7) are the faster $\ell 1$ electrons with longer lifetimes. We conclude that the mobility, which is an average over all electrons as in equation (A7), is dominated by contributions from electrons in the transverse ellipsoids. For each type of electron, however, the distinct phonon configurations that give rise to the ImΣ's plotted in figure 2 are *not* independent scattering mechanisms to be added in the usual way by adding their rates. Instead, they represent the range of fluctuations around the mean-field virtual configuration of partial occupancies prescribed by the Bose–Einstein distribution function. These fluctuations are not spatially homogeneous. In fact, figure 2 shows that, for $\ell 1$ electrons, fluctuations that produce large ImΣ are rare. We conclude that spatial fluctuations of phonon configurations create patches of low mobility and percolation 'paths of least resistance' for $\ell 1$ electrons in the transverse ellipsoids. Accordingly, we model the spatial fluctuations with a resistor network. Due to the overwhelming weight of the fast $\ell 1$ electrons, the network of high-conductivity regions is above the percolation threshold, so that the high-resistivity regions are 'islands' in this network. We can, therefore, approximate the averaged mobility by a parallel-resistor model, with each phonon configuration contributing to an independent conduction path so that the lifetime, rather than the scattering rate, is averaged. More

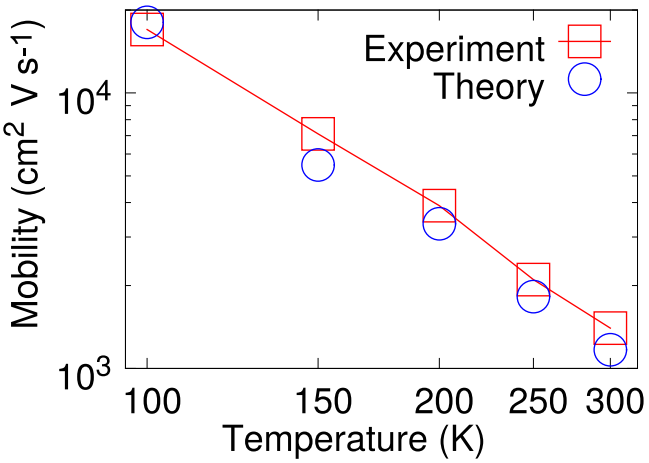

Figure 3. Electron mobility limited by phonon scattering as a function of temperature, theory (blue circles) compared to experiment [32] (red squares).

specifically, the mobility is calculated using

$$\mu = \frac{1}{N}\sum_{i=1}^{N}\frac{e}{12}\sum_{j=1}^{3}w_j\frac{\tau_{\ell j}\left(k_{x\ell j},\, k_{y\ell j},\, \frac{1}{2}k_B T\right)}{m_{z,\ell j}}, \qquad (6)$$

where N is the number of configurations and w_j is the weight, which is 8, 2 and 2 for $\ell 1$, $\ell 2$, $\ell 3$, respectively.

Figure 3 shows the calculated mobility as a function of temperature that experimental data [32] is in good agreement with. The theory is less accurate at low temperatures because of the rather poor representation of long-wave-length phonon modes in a finite-size supercell calculation. The results at low temperatures are not as well converged as those at higher temperatures. For comparison, if we were to treat each random phonon configuration as an independent scattering mechanism, the phonon-limited mobilities would be 20 to 30 percent lower.

The electron mobility limited by ionized-impurity scattering was calculated for P and As, with and without lattice relaxation, and for two different concentrations. One impurity out of 216 (512) atoms is equivalent to an impurity concentration of 2.26×10^{20} cm^{-3} (9.53×10^{19} cm^{-3}). We found that the result is very sensitive to screening by the electrons released by the impurities. Each impurity releases an electron to the conduction band and the whole system is neutral overall, whereby (on average) we must have one conduction-band electron in each supercell, i.e. the supercell is neutral. Thus, the Coulomb tail of the impurity potential, namely the difference between the electrostatic potentials in the supercell with an impurity at its center and in the pure supercell, must vanish at the edge of the supercell. We ensure that this is the case by aligning the electrostatic potential at the edge of the supercell with the impurity with the electrostatic potential at the edge of the pure supercell.

The calculated mobilities are summarized in table 1 and compared to experimental data [33]. There is good agreement for both dopants and for both concentrations. Lattice relaxation has a modest effect on the mobility.

Table 1. Electron mobility limited by ionized-impurity scattering and theory compared to experiment [33].

	Theory				Experiment [33]
2.26×10^{20} cm^{-3}	$\ell 1$	$\ell 2$	$\ell 3$	Total	
P (unrelaxed)	77.5	16.2	17.1	57.2	60
P (relaxed)	74.5	16.1	16.8	55.1	
As (unrelaxed)	53.3	12.4	12.1	39.5	45
As (relaxed)	52.7	11.5	12.1	39.2	
9.53×10^{19} cm^{-3}	$\ell 1$	$\ell 2$	$\ell 3$	Total	
P (unrelaxed)	111.7	28.4	29.8	84.2	80
P (relaxed)	106.3	28.4	28.7	80.2	
As (unrelaxed)	91.9	24.0	22.7	68.9	67
As (relaxed)	87.8	22.7	23.3	66.3	

In conclusion, we have developed a parameter-free method for the calculation of carrier mobilities in a semiconductor material that does not depend on the usual Born approximation. We demonstrated that fluctuations in phonon configurations, which are beyond the first-order Born approximation, in fact, dominate mobility by creating paths of least resistance. It is not clear how many orders in the Born expansion would be needed to capture this effect. Independent support for the existence of paths of least resistance is provided by noise measurements in Si channels showing levels of noise that are lower than predicted by electron-density fluctuations [34]. We also demonstrated that in ionized-impurity scattering, the Coulomb tail is cut off at roughly half the inter-impurity separation by screening. These results challenge the traditional assumptions that are described in all standard semiconductor monographs. Although in this paper we reported results on Si, the prototype semiconductor where the results can be unambiguously validated, they are relevant for mobilities in new semiconductors such as two-dimensional materials and complex oxides, all of which are considered for applications in electronics. Finally, the new method is faster than the Born-approximation methods and may prove useful in engineering applications for cases when predictions based on the Born approximation fail. The present results impact both fundamental understanding and the foundations of engineering-level modeling.

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Appendix A. Derivation of the mobility formula for silicon

The mobility is given within the linearized Boltzmann theory by

$$\mu = -\frac{2e}{n_c}\sum_j \int \mathrm{d}^3\mathbf{k}\frac{\tau_j(\mathbf{k})}{(2\pi)^3}v_j^2(\mathbf{k})\frac{\partial f(\epsilon_{\mathbf{k}j})}{\partial \epsilon_{\mathbf{k}j}}, \tag{A1}$$

where $\tau_j(\mathbf{k})$ is the scattering lifetime of the Bloch state with wavevector k and band index j. We use this formula separately for phonon and impurity scattering because they are independent mechanisms and their contributions to mobility can be added by Matthiessen's rule. In the case of impurity scattering, $\tau_j(\mathbf{k})$ is the elastic lifetime (which is equal to the momentum relaxation time if scattering is isotropic) [18]. For phonon scattering, it is the lifetime due to electron–phonon scattering, which is also elastic within the frozen phonon approximation. $v_j(\mathbf{k})$ is the group velocity, f is the Fermi–Dirac distribution function, $\epsilon_{\mathbf{k}j}$ is the band energy, and n_c is the density of carriers,

$$n_c = \frac{2}{(2\pi)^3}\sum_j \int \mathrm{d}^3\mathbf{k} f(\epsilon_{\mathbf{k}j}), \tag{A2}$$

with a factor of 2 to account for electron spin. It is convenient to separate out the direction of the current, which we define as the z direction. We use $dk_x dk_y dk_z = \sum_\ell dk_x dk_y \mathrm{d}\epsilon/\hbar v_{z,\ell}$ where ℓ counts the number of different k_z values for a given ϵ and $v_{z,\ell}$ is the group velocity along the z direction for electrons in valley ℓ. There would be two different k_z values for a single band. In the case of silicon, however, the conduction band has six equivalent minima so there are six different valleys, or a total of 12 terms counted by the index ℓ. We obtain,

$$\mu = -\frac{e\int \mathrm{d}k_x \mathrm{d}k_y d\epsilon \sum_\ell \tau_\ell(k_x, k_y, \epsilon) v_{z,\ell}(k_x, k_y, \epsilon)\frac{\partial f}{\partial \epsilon}}{\int \mathrm{d}k_x \mathrm{d}k_y d\epsilon \sum_\ell \frac{f(\epsilon)}{v_{z,\ell}(k_x, k_y, \epsilon)}}. \tag{A3}$$

Integrating by parts, we find,

$$\mu = -\frac{e\int \mathrm{d}k_x \mathrm{d}k_y d\epsilon \sum_\ell \frac{\tau_\ell(k_x, k_y, \epsilon)}{m_\ell v_{z,\ell}(k_x, k_y, \epsilon)} f(\epsilon)}{\int \mathrm{d}k_x \mathrm{d}k_y d\epsilon \sum_\ell \frac{f(\epsilon)}{v_{z,\ell}(k_x, k_y, \epsilon)}}. \tag{A4}$$

where we invoke the effective mass approximation for $\partial v_{z,\ell}/\partial\epsilon = 1/m_{z,\ell}v_{z,\ell}$ with $m_{z,\ell}$ as the effective mass along the z direction for valley ℓ, and we also neglected a term involving $\partial\tau_\ell/\partial\epsilon$. The scattering lifetime τ_ℓ depends weakly on k_x and k_y so we replace it with the value at the k point, $k_x = k_{x\ell}$ and $k_y = k_{y\ell}$, where the conduction band bottom is located (but note that τ_ℓ is not evaluated at the energy of the conduction band bottom). This allows the integration over $\mathrm{d}k_x \mathrm{d}k_y$ to be evaluated within the effective mass approximation,

$$\int \frac{\mathrm{d}k_x \mathrm{d}k_y}{v_{z,\ell}} = \frac{\pi}{\hbar^2}\sqrt{2m_{x,\ell}m_{y,\ell}m_{z,\ell}\epsilon}, \tag{A5}$$

where $m_{x,\ell}$ and $m_{y,\ell}$ are the effective masses along the x and y directions, respectively, for valley ℓ. This result is the same for all valleys because of symmetry. Therefore, we find,

$$\mu = -\frac{e\int \mathrm{d}\epsilon \sum_\ell \frac{\tau_\ell(k_{x\ell}, k_{y\ell}, \epsilon)}{m_{z,\ell}}\sqrt{\epsilon} f(\epsilon)}{12\int \mathrm{d}\epsilon \sqrt{\epsilon} f(\epsilon)}, \tag{A6}$$

where the factor of 12 in the denominator accounts for the six valleys in the silicon conduction band and two values of k_z for each valley (one left-going and one right-going). Finally, noting that $\epsilon - E_f \gg k_B T$, the factor, $\sqrt{\epsilon} f(\epsilon)$, in the integrand peaks at $\epsilon \approx \frac{1}{2}k_B T$, and that τ_ℓ depends weakly on the energy (the variation of τ_ℓ with respect to different defect configurations is much greater, as evident in figure 2), we replace $\tau_\ell(k_{x\ell}, k_{y\ell}, \epsilon)$ by its value, $\tau_\ell\left(k_{x\ell}, k_{y\ell}, \frac{1}{2}k_B T\right)$, at the peak of the rest of the integrand. This allows us to further simplify the mobility expression as,

$$\mu = \frac{e}{12}\sum_\ell \frac{\tau_\ell\left(k_{x\ell}, k_{y\ell}, \frac{1}{2}k_B T\right)}{m_{z,\ell}}. \tag{A7}$$

Appendix B. Calculation of phonon configurations

A set of frozen phonon configurations is generated with the displacement for each atom in a supercell given by,

$$\mathbf{u}_n = \sum_{\mathbf{q},i}\mathbf{u}_{n,\mathbf{q}i} = \frac{1}{\sqrt{m_n}}\sum_{\mathbf{q},i} A_{\mathbf{q}i}(T)\hat{n}_{\mathbf{q}i}\exp(i(\mathbf{q}\cdot\mathbf{R}_n + \phi_{\mathbf{q}i})), \tag{B1}$$

where m_n is the mass of atom n, $\mathbf{q}$ and i are mode indices, and $\hat{n}_{\mathbf{q}i}$ is the eigenvector for each mode and is normalized over the supercell, $\sum_n |\hat{n}_{\mathbf{q}i}|^2 = 1$. A random phase $\phi_{\mathbf{q}i}$ is added to remove correlation between different modes. $A_{\mathbf{q}i}(T)$ is a scaled temperature-dependent amplitude that represents the average effect of phonon motion. It is determined by considering the displacement operator,

$$\hat{d}_{\mathbf{q}i} = \sqrt{\frac{\hbar}{2\omega_{\mathbf{q}i}}}(\hat{b}_{\mathbf{q}i} + \hat{b}^\dagger_{\mathbf{q}i}). \tag{B2}$$

The thermally averaged value of $\langle\hat{d}^2\rangle$ is,

$$\langle\hat{d}^2_{\mathbf{q}i}\rangle = \frac{\hbar}{\omega_{\mathbf{q}i}}\left(N_{\mathbf{q}i} + \frac{1}{2}\right). \tag{B3}$$

where $N_{\mathbf{q}i} = \langle\hat{b}^\dagger_{\mathbf{q}i}\hat{b}_{\mathbf{q}i}\rangle$ is the average number of phonons for mode $\mathbf{q}i$ and is given by the Bose–Einstein distribution function,

$$N_{\mathbf{q}i} = \frac{1}{\exp(\hbar\omega_{\mathbf{q}i}/k_B T) - 1}. \tag{B4}$$

Setting $A^2_{\mathbf{q}i}(T) = \langle\hat{d}^2_{\mathbf{q}i}\rangle$ and noting $N_{\mathbf{q}i} \gg 1$, we find,

$$A_{\mathbf{q}i}(T) = \sqrt{\frac{N_{\mathbf{q}i}\hbar}{\omega_{\mathbf{q}i}}}. \tag{B5}$$

For each phonon configuration, self-consistent DFT calculation is performed to produce a Kohn–Sham potential, which is used to construct the effective medium.

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